

Yuk Tung Samuel Fang

Also published as: **Yudong Fang**

Nanjing University (Suzhou Campus) — Hong Kong SAR — B.Sc. Intelligent Science and Technology (Exp. 2027)

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Research Interests

Embodied AI and robotics, with a focus on perception, spatial memory, and decision-making for reliable action in unstructured real-world environments. Experience in navigation, semantic mapping, and real-robot deployment.

Education

Nanjing University (Suzhou Campus)

Suzhou, China

B.Sc. Intelligent Science and Technology

Expected 2027

- **GPA:** 4.47/5.00 — **Major GPA:** 4.45/5.00

- **Selected coursework:** Data Mining (95), Database Systems (96), Operating Systems (95), Data Structures & Algorithms (95), Machine Learning (94), Deep Learning (94), Programming Practicum (94.9)

Publications & Preprints

- **Fang, Y. T. S.**; Shi, Z.; Qiu, J.; Chen, Z.; Shi, J.; Xu, H.; Huo, J.; Gao, Y.

INHerit-SG: Incremental Hierarchical Semantic Scene Graphs with RAG-Style Retrieval.

ICRA 2026 Workshop on Robots Meet Prior Maps, Oral Presentation, Best Presentation Finalist.

arXiv:2602.12971 — Project Page

Contribution: First author and project lead; led the end-to-end system development, experiment design, manuscript preparation, and real-robot evaluation.

- Ding, H.; Xu, Z.; **Fang, Y. (Yudong)**; Wu, Y.; Chen, Z.; Shi, J.; Huo, J.; Zhang, Y.; Gao, Y.

LaViRA: Language-Vision-Robot Actions Translation for Zero-Shot Vision Language Navigation in Continuous Environments.

IEEE International Conference on Robotics and Automation (ICRA), 2026. arXiv:2510.19655 — Project Page

Contribution: Led real-robot deployment on Unitree Go1 and system integration.

Research Experience

Research Intern, National University of Singapore (NUS)

2026.07–present

Advisor: Prof. Lin Shao

- Working onsite on long-horizon mobile manipulation and embodied agent evaluation.

Undergraduate Researcher, Nanjing University

2025–present

Inference & Learning Research Group (led by Prof. Yang Gao), State Key Laboratory for Novel Software Technology

Advisor: Prof. Jieqi Shi

Incremental Hierarchical Semantic Scene Graphs (INHerit-SG)

2025.10–2026.02

Advisors: Prof. Jieqi Shi, Prof. Hao Xu, Prof. Jing Huo

- First author and project lead; led the end-to-end system development, experiment design, manuscript preparation, and real-robot evaluation.

- Introduced an online Floor-Room-Area-Object hierarchy with RAG-style retrieval, natural-language semantic anchors, and event-triggered updates for long-term consistency.

- Constructed HM3DSem-SQR and a human-study dataset; achieved 37.7% geometric accuracy (within 1 m), 70.6% semantic accuracy, and 60.0% real-world trajectory success.

Zero-Shot Vision-and-Language Navigation (LaViRA)

2025.05–2025.09

Advisors: Prof. Jieqi Shi, Prof. Jing Huo

- Developed a coarse-to-fine hierarchy that translates language plans into visually grounded actions and precise robot controls for zero-shot VLN-CE.

- Achieved 38.3% Success Rate and 28.3% SPL with Gemini-2.5-Pro; led real-robot deployment and system integration on Unitree Go1 and AgileX Cobot Magic.

Research Experience (continued)

Active Exploration with Semantic Map Prediction (SEA)

2025.09–2025.10

Advisors: Prof. Jieqi Shi, Prof. Jing Huo

- Developed an iterative prediction-exploration framework with confidence-aware semantic map updates and hierarchical navigation policies.
- Deployed and debugged the full Habitat pipeline with real-robot integration on Unitree Go1 and AgileX Cobot Magic.

Research in Progress

Lifelong Mobile-Manipulation Memory Benchmark

2026.03–present

- Designing a benchmark for world-model maintenance under continuous household perturbations, covering model construction, conflict-aware updates, and memory-guided action; simulation environment implemented.

Uncertainty-Aware Interactive Object-Goal Navigation

2026.06–present

- Developing a framework for deciding when an agent should act, explore, or ask the user using uncertainty modeling and information gain; validated feasibility through baseline reproduction.

Honors & Awards

- NJU Scholarship for HK/Macao & Overseas Chinese Students — **First Prize** (Sophomore year; university-wide; 5 awardees, Top 2%; defense required)
- NJU Scholarship for HK/Macao & Overseas Chinese Students — **Third Prize** (Freshman year; university-wide)

Academic Service

- Reviewer (by invitation), *IEEE Robotics and Automation Letters* (RA-L), 2026

Skills

- **Programming:** Python, C++ — **Frameworks & Tools:** PyTorch, Linux, Git
- **Robotics & Simulation:** ROS, Habitat, AI2-THOR/ProcTHOR, real-robot deployment and debugging
- **Hardware:** Unitree Go1, AgileX Cobot Magic, Jetson AGX Orin, RealSense D435i, Gemini 336L, Manifold Odin 1; 3D modeling and printing